Binary Logistic Regression for Predicting Hazardous Asteroids

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1 Introduction

In recent years, the rapid growth of data across scientific fields has increased the importance of statistical modeling and machine learning methods. Regression analysis plays a central role in understanding relationships between variables and making predictions [2].

When the response variable is binary, linear regression is not appropriate. Logistic regression provides a probabilistic framework for classification problems and is widely used for modeling binary outcomes [4].

Statistical learning methods aim to extract meaningful patterns from data while balancing interpretability and prediction accuracy [3].

In this study, binary logistic regression is used to predict whether an asteroid is hazardous using data obtained from NASA.

2 Literature Review

Regression methods are fundamental tools in statistical analysis [2]. Logistic regression, in particular, is widely used for classification problems [5].

Hastie et al. (2023) emphasize the importance of classification models in statistical learning and highlight logistic regression as a key method [3].

James et al. (2023) provide a practical introduction and show how logistic regression is used in real-world applications [4].

3 Research Objective

The objective of this study is to develop a binary logistic regression model to predict whether an asteroid is potentially hazardous.

Specifically, this study aims to:

- Identify significant predictors of asteroid hazard
- Interpret the effect of variables on log-odds
- Evaluate model performance using classification metrics

4 WHY LOGISTIC

Binary logistic regression was selected in this study because the response variable is binary, indicating whether an asteroid is hazardous or not. Linear regression is not suitable for such problems, as it can produce predictions outside the interval [0,1] and does not model probabilities appropriately.

Logistic regression, on the other hand, provides a probabilistic framework that ensures predicted values remain between 0 and 1. In addition, it allows interpretation of the relationship between predictors and the log-odds of the outcome, making it both practical and interpretable for classification problems.

5 Theoretical Background

Unlike linear regression, logistic regression is specifically designed for binary outcomes. Linear regression assumes a continuous response and may produce invalid probability estimates when applied to classification problems.

Logistic regression overcomes this limitation by using the logistic function to constrain predicted values between 0 and 1. This makes it more appropriate for modeling probabilities and classification tasks.

In linear regression, parameters are typically estimated using Ordinary Least Squares (OLS), which minimizes the sum of squared errors. However, OLS is not suitable for binary outcomes because it assumes normally distributed errors and can produce predicted values outside the interval [0,1].

In contrast, logistic regression uses Maximum Likelihood Estimation (MLE), which estimates parameters by maximizing the likelihood of observing the given data. Since the response variable follows a Bernoulli distribution, MLE is more appropriate and provides

consistent and efficient estimates for binary classification problems. This difference between OLS and MLE is one of the key reasons why logistic regression is preferred over linear regression for classification problems.

Logistic regression models the probability of a binary outcome:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}}$$

The logit transformation is:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Parameters are estimated using Maximum Likelihood Estimation (MLE).

6 Methodology

6.1 Model Assumptions

Binary logistic regression relies on several assumptions:

- Binary outcome variable
- Independence of observations
- No multicollinearity between predictors
- Linearity between predictors and log-odds
- Sufficient sample size

6.2 Dataset

The dataset was obtained from NASA's Near Earth Object API and contains 10 years of asteroid observations. The dataset includes 64,117 observations.

The target variable is `is_potentially_hazardous`, which indicates whether an asteroid is hazardous. The dataset includes several important physical characteristics of asteroids, such as absolute magnitude, estimated diameter, relative velocity, and miss distance. These variables are commonly used in astronomical studies to assess asteroid properties and potential risk.

The relatively large size of the dataset (over 64,000 observations) provides a strong basis for statistical modeling. However, the imbalance between hazardous and non-hazardous asteroids introduces challenges for classification, making careful model evaluation necessary.

6.3 Data Preprocessing

- Irrelevant variables (id, name, date) were removed
- Target variable converted to binary (0/1)
- No missing values were found
- Multicollinearity reduced by removing one diameter variable

6.4 Exploratory Data Analysis

The dataset is highly imbalanced:

- Non-hazardous: 94.7%
- Hazardous: 5.3%

Therefore, accuracy alone is not sufficient.
Hazardous asteroids tend to:

- have higher velocity
- have larger diameter

Distance shows weaker effect.

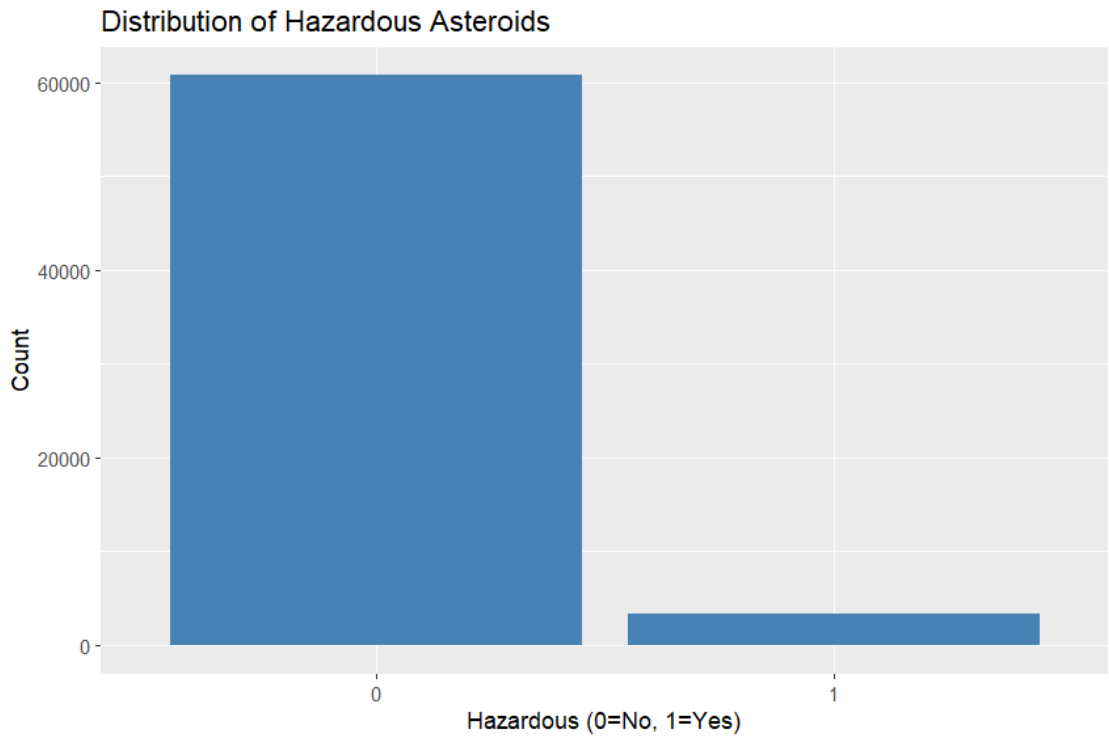

Figure 1: Class distribution

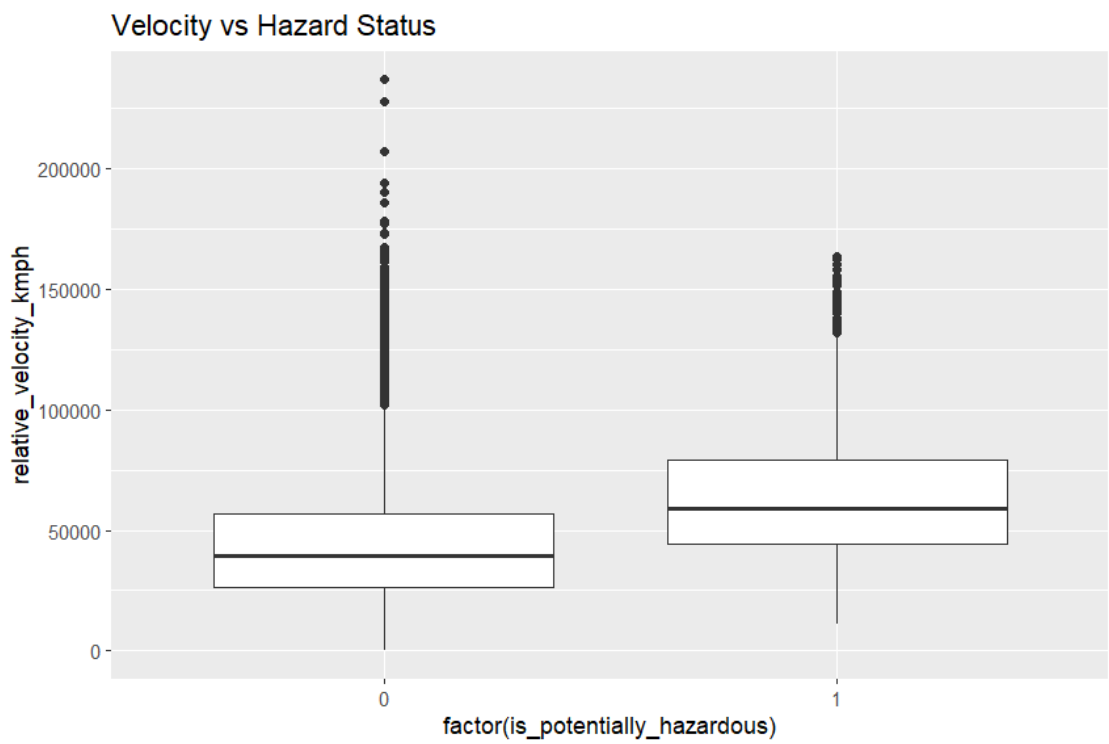

Figure 2: Velocity vs Hazard

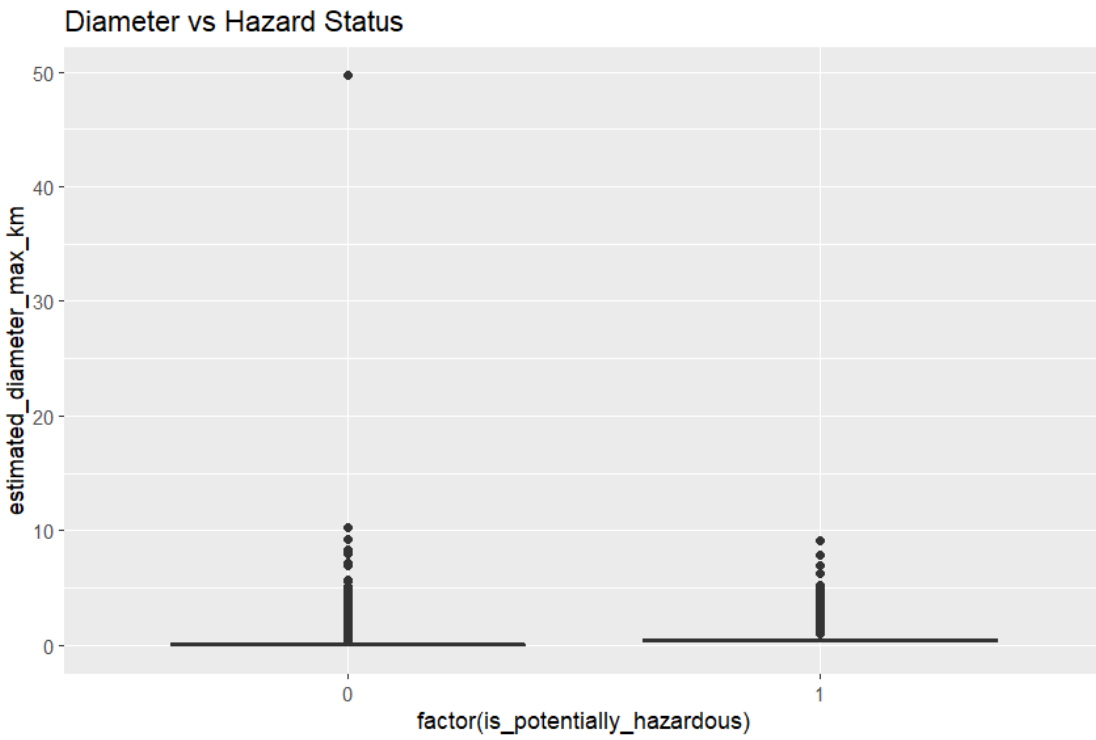

Figure 3: Diameter vs Hazard

6.5 Model Implementation

The logistic regression model was implemented in R using the `glm()` function.

6.6 Model Evaluation

The model was evaluated using:

- Confusion Matrix
- Accuracy
- Sensitivity
- Specificity
- ROC Curve and AUC

Due to class imbalance, accuracy is not sufficient alone.

7 Results

The results clearly show that the model performs well in terms of overall classification ability. The AUC value of 0.94 indicates that the model has a strong ability to distinguish between hazardous and non-hazardous asteroids.

However, when using the default threshold of 0.5, the model achieves high accuracy but fails to detect most hazardous asteroids. This is reflected in the very low sensitivity value.

After lowering the threshold to 0.2, the model becomes more effective in identifying hazardous asteroids. Although the overall accuracy slightly decreases, the improvement in sensitivity and balanced accuracy makes the model more suitable for practical applications.

These findings demonstrate that evaluating classification models requires more than accuracy, especially in imbalanced datasets.

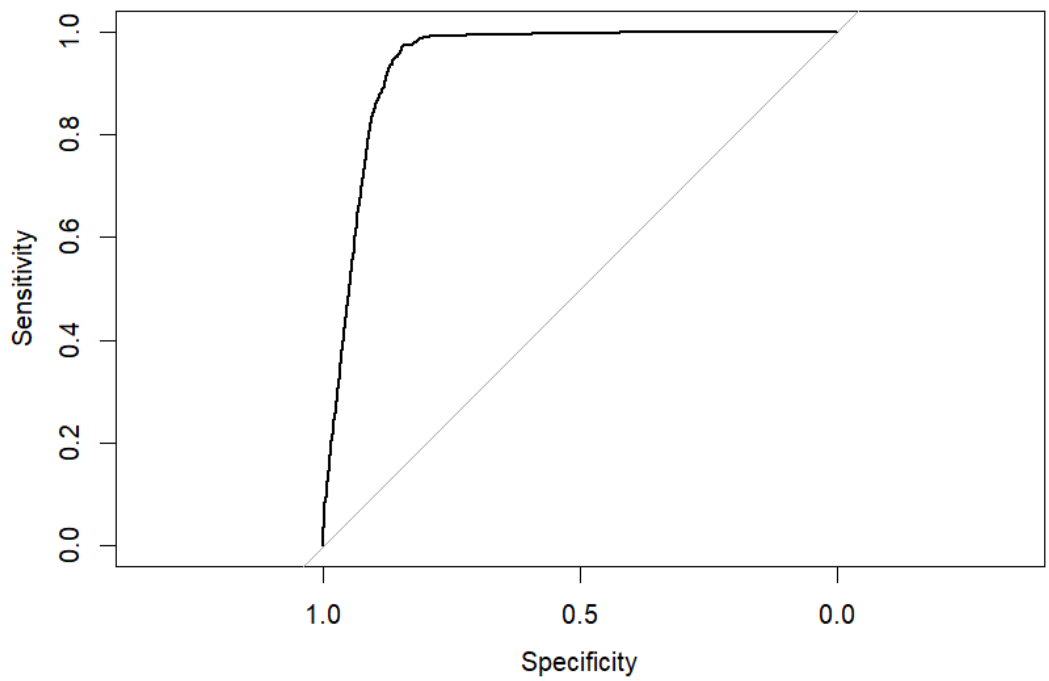

Figure 4: ROC Curve

7.1 Threshold Comparison

Threshold	Accuracy	Sensitivity	Specificity	Balanced Accuracy
0.5	0.946	0.13	0.99	0.56
0.2	0.915	0.67	0.93	0.80

Table 1: Model Performance Comparison

The table clearly shows that lowering the threshold improves sensitivity and balanced accuracy, making the model more effective for detecting hazardous asteroids.

At threshold 0.5, the model fails to detect hazardous asteroids.

At threshold 0.2, the model becomes balanced and more effective.

8 Discussion

The model demonstrates strong classification performance with an AUC value of 0.94.

However, due to class imbalance, default threshold gives misleading results.

Lowering the threshold improves sensitivity and makes the model more useful.

Velocity and size are key predictors of asteroid hazard.

The consistency between exploratory analysis and model results strengthens the validity of the findings. Variables such as velocity and size were identified as important both in visual analysis and in the logistic regression model.

This consistency suggests that the model captures meaningful relationships in the data, rather than producing results by chance.

Another important implication of this study is that real-world classification problems often require trade-offs between different performance metrics.

In this case, improving sensitivity was more important than maintaining maximum accuracy, as failing to detect hazardous asteroids may have serious consequences.

This highlights the importance of domain knowledge in model evaluation and decision-making.

8.1 Limitations

One limitation of this study is the class imbalance in the dataset, which may affect model performance.

In addition, logistic regression assumes a linear relationship between predictors and the logit, which may not fully capture complex relationships in the data. Future work may

include applying more advanced machine learning models to compare performance with logistic regression.

8.2 Future Work

Future research could explore more advanced machine learning models such as random forests, support vector machines, or neural networks to compare performance with logistic regression.

Additionally, techniques such as resampling or cost-sensitive learning could be applied to address class imbalance more effectively.

9 Conclusion

Binary logistic regression successfully models asteroid hazard using the given dataset.

The model demonstrates strong classification performance, as indicated by the AUC value. However, the results highlight that threshold selection is critical, especially in imbalanced datasets. A lower threshold significantly improves the model's ability to detect hazardous asteroids and enhances its practical usability.

This study highlights the importance of using appropriate evaluation metrics in imbalanced classification problems.

While accuracy may suggest strong performance, it can be misleading in cases where one class dominates the dataset.

The findings demonstrate that threshold tuning plays a critical role in improving model effectiveness.

Overall, logistic regression provides a simple yet powerful framework for classification tasks, especially when interpretability is important.

References

[1] NASA (2024). *Near Earth Object Web Service (NeoWs)*. Available at: `https://api.nasa.gov/`

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